

CPP Book of Questions

Paul Sherwood

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Chapter 1

1.1 Level 1

Question 1: `std::print`

What does `std::print` provide compared to `printf` or `std::cout`?

Answer 1.1.1: `std::print`

Type safety, checked at compile time. Can format custom types (with `std::formatter`) Uses python formatting Very concise and easy to use.

Question 2: `auto`

What is the advantage of `auto` and type deduction in modern C++?

Answer 1.1.2: `auto`

Prevent types mismatches.
Easy use for templates
clean syntax

Question 3: Passing by _

What is the difference between passing by value, reference, and const reference?

Answer 1.1.3: `auto`

value complete copy reference use a pointer to edit the original variable const ref used like a read only of the original variable

Question 4: bindings

What are structured bindings and when would you use them?

Answer 1.1.4: `auto`

Question 5: Ranges

What are C++23 ranges and views?

Answer 1.1.5: auto

Question 6: constexpr

What does constexpr mean and when is it evaluated?

Answer 1.1.6: auto

Question 7: Vector vs Arrays

What is the difference between std::vector and raw arrays?

Answer 1.1.7: auto

Question 8: string_view

What is std::string_view and why is it more efficient than std::string?

Answer 1.1.8

Question 9: ranges

What is the purpose of #include <<ranges>

Question 10: lifetime

What is the lifetime of a temporary returned from a function?